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Classifying multipartite continuous-variable entanglement structures through data-augmented neural networks

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Neural networks have emerged as a promising paradigm for quantum information processing, yet they confront the challenge of generating training datasets with sufficient size and rich diversity, which is particularly acute when dealing with multipartite quantum systems. For instance, in the task of classifying different structures of multipartite entanglement in continuous-variable systems, it is necessary to simulate a large number of infinite-dimensional state data that can cover as many types of non-Gaussian states as possible. Here we develop a data-augmented neural network to address this task with homodyne measurement data. A quantum data augmentation method based on classical data processing techniques and quantum physical principles is proposed to efficiently enhance network performance. By testing on randomly generated tripartite and quadripartite states, we demonstrate that the network can infer the entanglement structure among the various partitions, and the accuracies are substantially improved with data augmentation. Our approach allows us to further extend the use of data-driven machine learning techniques to more complex tasks of learning quantum systems encoded in a large Hilbert space.

Inspired by biological systems, neural networks have demonstrated remarkable success across diverse domains including quantum physics. These models offer innovative solutions to challenges such as quantum state reconstruction[1–3], feature learning[4–9] and even fundamental physics discovery[10–13]. However, the efficacy of these data-driven techniques—for both classical and quantum problems—critically depends on the quality of the underlying datasets[14–16]. Although increasing the size and diversity of training data enhances a network's ability to generalize to unseen cases, assembling such comprehensive datasets demands substantial time and resources for meticulous collection and annotation.

This becomes more challenging when dealing with multipartite quantum systems in which complexity escalates with the number of subsystems. Detecting multipartite entanglement exemplifies this difficulty. Multipartite entanglement, essential in quantum physics, is classified by its separability across all possible combinations of

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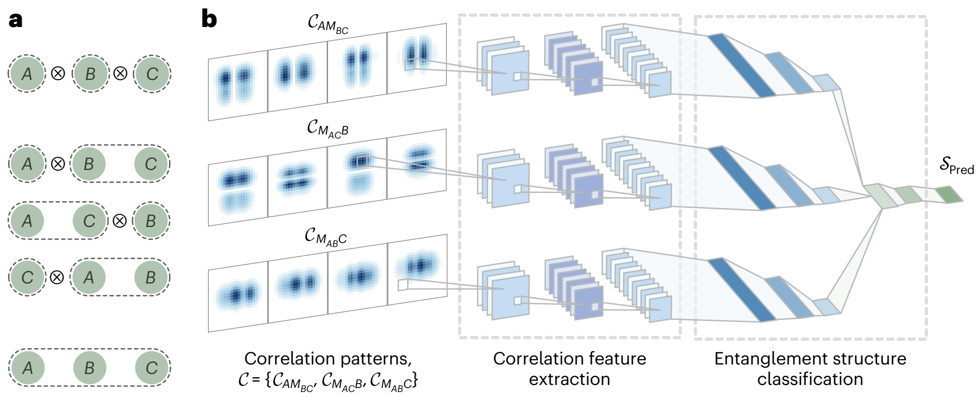

Fig. 1 | Multiclass classification task. a, Tripartite entanglement structures for the partitions of fully separable partition $1\otimes1\otimes1$, three biseparable partitions $1\otimes2$ and fully inseparable states. **b**, Neural network architecture for this task. Three groups of correlation patterns $\mathcal{C}_{AM_{BC}}$, $\mathcal{C}_{M_{AC}B}$ and $\mathcal{C}_{M_{AB}C}$ provide correlation features of $A-BC$, $B-AC$ and $C-AB$ splittings, respectively. $M_{ij}(i,j\in\{A,B,C\})$ denotes one of the output modes by mixing modes i and j on a balanced beamsplitter. This mixing strategy enables more effective extraction of information across different partitions and reducing the number of required measurements. The features are extracted by convolutional layers and subsequently classified by fully connected layers into three classes of tripartite entanglement structures as shown in **a**, resulting in a predicted entanglement structure label $\mathcal{S}_{\mathrm{Pred}}$.

subsystems, that is, multipartitions[17–20]. For some typical states, such as multipartite Gaussian states, entanglement conditions in terms of different multipartitions have been demonstrated[21]. However, classifying the entanglement structures of arbitrary multipartite non-Gaussian states, whose non-trivial correlations exist in higher-order moments, remains experimentally infeasible.

Efforts to harness neural networks for classifying entanglement structures in discrete-variable systems have been made[22–25], aiming to circumvent the high cost of full quantum state tomography[26]. Conceptually, this approach aligns with broader machine learning studies that extract global quantum properties from high-dimensional many-body data. For example, classical neural networks have been used to identify phases of matter from spin configurations and entanglement spectra[27,28], and quantum convolutional neural networks process quantum states directly through structured quantum circuits[29]. In contrast to these finite-dimensional many-body settings, the present work addresses multipartite entanglement structure classification in continuous-variable (CV) systems. In such systems, characterized by infinite-dimensional Hilbert spaces[30], conventional tomography becomes even less practical[31]. Neural networks have offered an experimentally viable alternative by analysing statistical features extracted from quadrature components of light through homodyne measurements, demonstrating success in bipartite settings[9]. For multipartite non-Gaussian states, however, the key difficulty lies not only in designing an effective classifier but also in overcoming the major obstacle of generating sufficiently diverse and reliably labelled training samples—a challenge that hinders the direct generalization of existing techniques to this setting.

The contributions of this work are twofold. First, we develop a neural network model capable of accurately classifying entanglement structures of arbitrary multipartite CV states using experimentally accessible homodyne measurement data as input. Second, to address the challenge in simulating multipartite CV systems, we propose a quantum data augmentation (QDA) method that substantially enhances the accuracy of predicting entanglement structures. Data augmentation (DA) refers to expanding datasets by transforming the given data into new, label-preserving samples. In the quantum context, these operations can be derived from quantum physical principles such as entanglement invariance under mode permutation and the convexity of separable states.

We apply the data-augmented neural network for tripartite and quadripartite CV systems in all possible multipartitions. The dataset covers states with varying degrees of non-Gaussian complexity, including highly non-Gaussian cases such as cat states. Compared with accuracies of 0.961 and 0.796 obtained using only original datasets for the two cases, the network with augmented data can remarkably improve them to 0.986 and 0.928, respectively. Therefore, our results provide an efficient approach for classifying multipartite non-Gaussian entanglement structures in an experimentally accessible way, which could exhibit metrological power in quantum phase estimation[32]. Moreover, inspired by the resource-preserving free operations in quantum resource theory[33], a wider range of label-preserving transformations can be designed as QDA operations. Hence, neural networks can be extended for broader tasks in learning multipartite quantum systems, even when data acquisition cost is exorbitant.

Results

Multipartite entanglement detection goes beyond simply identifying the presence of entanglement; it also involves determining its detailed structure. Specifically, any $\mathcal{R}_1|\cdots|\mathcal{R}_j$-separable state can be decomposed in the form $\hat{\rho}=\sum_\gamma p_\gamma \hat{\rho}^{(\gamma)}_{\mathcal{R}_1}\otimes\cdots\otimes\hat{\rho}^{(\gamma)}_{\mathcal{R}_j}$, where $\mathcal{R}_i$ describes an ensemble of modes and $\hat{\rho}^{(\gamma)}_{\mathcal{R}_i}$ denotes the reduced density matrix of $\hat{\rho}^{(\gamma)}$ on the subsystem $\mathcal{R}_i$ (ref. 34). In the following task, we classify cases with the same number of modes in each subsystem $\mathcal{R}_i$ as the same multipartition class and label them by $\mathcal{S}$. For example, states with separability $A|BC$, $B|AC$ and $C|AB$ belong to the multipartition class $\mathcal{S}=1\otimes2$.

Quantum state set construction

We train the neural network on a broad set of random, mixed CV states, including all Gaussian states and most experimentally achievable non-Gaussian states. As illustrated in Fig. 1a, we start with the simplest multipartite case: tripartite states. To collect balanced samples for all possible classes of states with different multipartitions $\mathcal{S}$ in a random way, we first generate a large number of m-mode ($m\le3$) seed states $\hat{\sigma}_m$. Then, we use these seed states to separately construct each class. For example, states with $\mathcal{S}=1\otimes1\otimes1$ can be decomposed into a tensor product of three single-mode seed states $\otimes_{i=1}^{3}\hat{\sigma}_1^{(i)}$, whereas states with $\mathcal{S}=1\otimes2$ can be decomposed into a tensor product $\hat{\sigma}_1\otimes\hat{\sigma}_2$ of a single-mode seed state $\hat{\sigma}_1$ and a two-mode entangled seed state $\hat{\sigma}_2$.

The seed states $\hat{\sigma}_m$ are generated based on the stellar formalism, which gives an operational characterization of non-Gaussian states through their stellar functions[35–37]. For a normalized pure state $|\psi\rangle$, the stellar function is defined as $F^\star_\psi(\mathbf{z})\equiv e^{\frac{1}{2}\|\mathbf{z}\|^2}\langle\mathbf{z}^*|\psi\rangle$, where $|\mathbf{z}\rangle$ denotes a coherent state with complex amplitude $\mathbf{z}=(z_1,\ldots,z_m)\in\mathbb{C}^m$.

Coherent states are the eigenstates of the annihilation operator and serve as the building blocks of CV quantum optics; their complex amplitudes parameterize displacements in phase space. The number of zeros of $F^{\star}_{\psi}(\mathbf{z})$ defines the stellar rank r, which quantifies the non-Gaussianity of state $|\psi\rangle$ and represents the minimal number of non-Gaussian operations required to engineer the state from a vacuum. Gaussian states, including coherent, squeezed and thermal states, are formally characterized by Gaussian characteristic functions and quasi-probability distributions on the multimode phase space; consequently, all Gaussian states have $r = 0$, and the stellar rank is preserved under all Gaussian unitaries. This formalism, thus, provides a rigorous yet accessible way to classify non-Gaussian resources in CV quantum systems.

We focus on states with $r \in \{0, 1, 2, 3, +\infty\}$, which include both finite- and infinite-stellar-rank states. For the finite case, any m-mode pure state can be decomposed into $\hat{G}|C(r)\rangle$, where $\hat{G}$ is a general m-mode Gaussian unitary consisting of squeezing, displacement and beamsplitters, and $|C(r)\rangle$, called a core state[36], is an m-mode state with finite support over the Fock basis and stellar rank r. To summarize, each m-mode seed state $\hat{\sigma}_m$ is obtained from a random core state $|C(r)\rangle$ with a given finite stellar rank r, followed by an m-mode Gaussian unitary with random parameters and m loss channels[38] with random efficiencies η_l ($l = 1...m$), given by

$$\hat{\sigma}_m = \left(\prod_{l=1}^{m} \hat{L}_l(\eta_l)\right) \hat{G}|C(r)\rangle\langle C(r)|\hat{G}^{\dagger} \left(\prod_{l=1}^{m} \hat{L}_l^{\dagger}(\eta_l)\right), \quad (1)$$

where each $\hat{L}_l$ is a Kraus operator of the loss channel[39,40]. For the infinite case, seed states consist of multimode entangled cat states and single-mode coherent states, subjected to the same loss channels (Supplementary Section IA,B).

Following the definition of an entanglement structure in terms of multipartitions, entangled seed states are required as necessary components in these cases, except for fully separable partitions. Furthermore, these entangled seed states must be fully inseparable to rule out any possible separability from seed states and avoid overlap between different multipartite separability classes. Here we adopt a recently proposed criterion based on quantum Fisher information (QFI) to identify these fully inseparable seed states[41]. The QFI-based criterion detects multipartite entanglement by systematically optimizing a local operator $\hat{A}$ and testing for violation of inequality between QFI and corresponding variances as $F_Q(\hat{\rho}_k, \hat{A}) \le 4V(\hat{\rho}_k, \hat{A})$, which holds for all k-separable states. However, applying this criterion to the detection of multipartite non-Gaussian entanglement fundamentally requires complete knowledge of the density matrix, rendering it infeasible for direct implementation in practical experiments. Nevertheless, we can apply it to the theoretically simulated density matrices in the training set to generate reliable entanglement labels. In the tripartite case, we randomly generate a dataset of 30,000 states, with 10,000 states for each entanglement class, including the fully separable class.

Correlation patterns from homodyne detection

We aim to map certain experimentally feasible correlation features, for example, patterns extracted from homodyne detection, to multipartite entanglement structures described by different multipartitions $\mathcal{S}$. Homodyne detection is a projective measurement on eigenstates of orthogonal quadrature operators $\hat{x}_l = (\hat{a}_l^{\dagger} + \hat{a}_l)$ and $\hat{p}_l = \mathrm{i}(\hat{a}_l^{\dagger} - \hat{a}_l)$, where $\hat{a}_l$ is the photon annihilation operator of mode $l \in \{A, B, C\}$ (refs. 31,42). According to Born's rule, with $\hat{x}_l|X_l\rangle = X_l|X_l\rangle$, the probability of obtaining the outcome (X_A, X_B, X_C) when simultaneously measuring $\hat{x}_l$ on the three modes of state $\hat{\rho}$ is given by $\mathcal{P}(X_A, X_B, X_C) \equiv \langle X_A; X_B; X_C|\hat{\rho}|X_A; X_B; X_C\rangle$. To ensure correlation features are not overlooked when only measuring reduced states, we use a balanced beamsplitter $\hat{\mathcal{U}}_{ij} = \mathrm{e}^{\frac{\pi}{4}(\hat{a}_i^{\dagger}\hat{a}_j - \hat{a}_i\hat{a}_j^{\dagger})}$ to mix modes i and j, producing two output modes M_{ij} and N_{ij}. Measuring M_{ij} and the remaining unmixed mode k yields two-variable quadrature statistics, thereby extracting more pattern information with fewer measurements. As an example, by mixing modes B and C and then measuring quadrature X_A of mode A and quadrature $X_{M_{BC}}$ of the mixed mode M_{BC}, the joint probability is given by

$$\begin{aligned}\mathcal{P}(X_A, X_{M_{BC}}) &= \langle X_A; X_B|\hat{\mathcal{U}}_{BC}\hat{\rho}\,\hat{\mathcal{U}}_{BC}^{\dagger}|X_A; X_B\rangle \\ &= \langle X_A; \tfrac{X_B+X_C}{\sqrt{2}}|\hat{\rho}|X_A; \tfrac{X_B+X_C}{\sqrt{2}}\rangle.\end{aligned} \quad (2)$$

Apart from $\mathcal{P}(X_A, X_{M_{BC}})$, we simulate the other three joint probability distributions of $\mathcal{P}(X_A, P_{M_{BC}})$, $\mathcal{P}(P_A, X_{M_{BC}})$ and $\mathcal{P}(P_A, P_{M_{BC}})$ to capture correlation features between subsystems A and BC, which corresponds to one of the bipartitions we aim to detect (Supplementary Fig. 5). Considering the other two cases with respect to $B-AC$ and $C-AB$ splittings, we simulate a total of 12 joint probability distributions for each state. To process them through the network, we bin every joint probability distribution into a 24 × 24-pixel^2 grid, named a correlation pattern. Although this discretization unavoidably causes some loss of information, the neural network exhibits strong robustness against such degradation (see the 'Discussion' section). In addition, to account for statistical fluctuations inherent in real experiments, we evaluate the network on datasets simulated from finite sampling points using the Monte Carlo algorithm, which is also given in the 'Discussion' section.

Network input and training strategy

For each tripartite state $\hat{\rho}$, we compute 12 correlation patterns, organized into three groups $\mathcal{C} = \{\mathcal{C}_{AM_{BC}}, \mathcal{C}_{M_{AC}B}, \mathcal{C}_{M_{AB}C}\}$. Each group contains four distinct patterns derived from homodyne measurements. Importantly, these patterns, which represent binned joint probability distributions of quadrature measurements (equation (2)) rather than mere second-order correlations, serve as the input to the network. The corresponding ground-truth multipartitions $\mathcal{S}_{\mathrm{True}}$, rigorously determined from the full density matrix of each state, serve as the output labels during supervised training.

The network architecture (Fig. 1b) is composed of three parallel convolutional sub-networks[43]. Each sub-network independently processes one group of four correlation patterns and condenses them into a feature vector. The three vectors are then combined through fully connected layers, and classified to a final outcome, that is, the predicted multipartition class $\mathcal{S}_{\mathrm{Pred}}$. We randomly hold 80% of the 30,000 simulated samples as an original training dataset, with the loss function being categorical cross-entropy (Supplementary Fig. 11).

During the supervised training phase, the network learns a complex, nonlinear mapping from experimentally accessible quadrature data to the entanglement properties defined by the full density matrix. Once trained, it can accurately infer non-Gaussian entanglement directly from such simple inputs in the testing phase. This design bypasses the resource-intensive full state tomography traditionally required for certifying non-Gaussian entanglement, and captures entanglement structures that remain inaccessible to criteria based purely on second-order moments.

QDA

The performance of neural networks heavily depends on the size and diversity of training data. However, for a multipartite non-Gaussian system, the data complexity increases with the number of modes m and stellar rank r (ref. 44), demanding larger datasets to capture this growing diversity. Therefore, efficiently producing comprehensive training data for multipartite CV systems is central to addressing these machine learning problems. To solve this, we adopt QDA, a technique originally developed in classical tasks to synthesize samples from an existing dataset, thereby efficiently enhancing model performance.

The augmentation operations typically involve subtle variations that do not affect the model's predictions. An example of classical DA via image flipping is shown in Fig. 2a, where a cat flipped horizontally remains a cat, preserving its label in an image classification task[45].

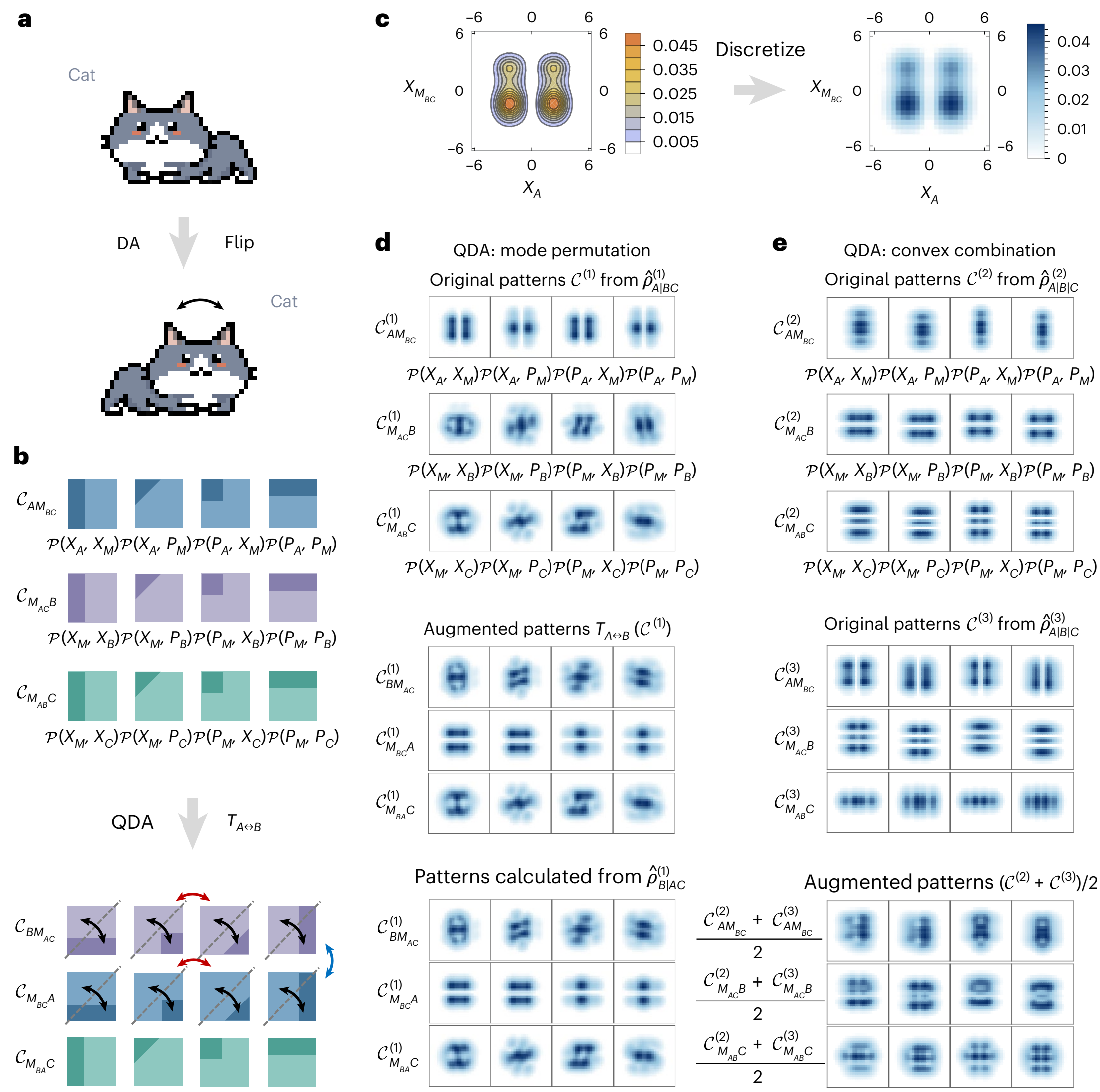

Fig. 2 | Possible strategies of QDA on the input correlation patterns. a, In the classical image augmentation process, flipping creates new samples without altering the semantic label 'cat'. **b**, Schematic of how correlation patterns $\mathcal{C}$ are transformed when modes A and B are swapped, and maintaining both validity of transformed quantum measurement data and invariance of the multipartitions label $\mathcal{S}$. Black arrows represent flipping along the dashed lines, red arrows represent exchanging the patterns and the blue arrow represents swapping two rows. **c**, Left: theoretical joint probability distribution $\mathcal{P}(X_A, X_{M_{BC}})$ of a tripartite non-Gaussian state (equation (2)). Right: the corresponding correlation pattern obtained by binning the distribution shown on the left. **d**, Correlation patterns $\mathcal{C}^{(1)}$ of an original $A|BC$ biseparable state $\hat{\rho}^{(1)}_{A|BC}$ (top). By rearranging $\mathcal{C}^{(1)}$ according to the mode-permutation transformation $T_{A\leftrightarrow B}$ illustrated in **b**, QDA efficiently produces the corresponding patterns for the $B|AC$ partition at a negligible time cost (middle). The same patterns can be obtained by explicitly constructing the permuted state $\hat{\rho}^{(1)}_{B|AC}$ and then recalculating the correlation patterns from its density matrix (bottom), but this direct simulation route is substantially more time-consuming. **e**, Correlation patterns $\mathcal{C}^{(2)}$ and $\mathcal{C}^{(3)}$ of two distinct fully separable states $\hat{\rho}^{(2)}_{A|B|C}$ and $\hat{\rho}^{(3)}_{A|B|C}$ (top and middle). A new sample of patterns can be generated via QDA through the convex combination $(\mathcal{C}^{(2)}+\mathcal{C}^{(3)})/2$ (bottom).

Inspired by the success of DA in classical tasks, we introduce a QDA method and explore how augmented data can be constructed for learning quantum systems. Similar to classical tasks, these augmentation operations preserve the labels of quantum states.

The first QDA operation is based on mode permutation, denoted as T. Since the entanglement structure is invariant with respect to mode indexing, the label $\mathcal{S}$ remains unchanged when rearranging the modes of state $\hat{\rho}$. An example of this operation is given in Fig. 2b. The mode permutation $T_{A\leftrightarrow B}$ can be easily achieved by flipping the pattern diagonally and then exchanging correlation patterns in the first two rows and the middle two columns. Figure 2d (top) shows the simulated correlation patterns $\mathcal{C}^{(1)}$ of a biseparable state $\hat{\rho}^{(1)}_{A|BC}$. The middle panel shows the QDA-transformed patterns $T_{A\leftrightarrow B}(\mathcal{C}^{(1)})$, obtained by applying the mode-permutation rule (Fig. 2b) directly to the correlation patterns. This transformation preserves the multipartition label $\mathcal{S}=1\otimes 2$ and produces a distinct training sample for the network. As a reference, the bottom panel shows the patterns obtained by explicitly permuting modes A and B at the density-matrix level and then calculating the corresponding patterns. The two routes produce the same transformed patterns, but QDA avoids the costly density-matrix recalculation (Supplementary Table 1) and obtains them through a simple rearrangement, thereby substantially reducing the sample generation time. In principle, for an m-mode quantum system, QDA based on mode permutation can expand the size of the dataset to $m!$ times its original size.

Apart from this, some quantum properties can also be used to define the QDA operations. For instance, in our task, a convex combination of any fully separable states remains separable[19], thereby

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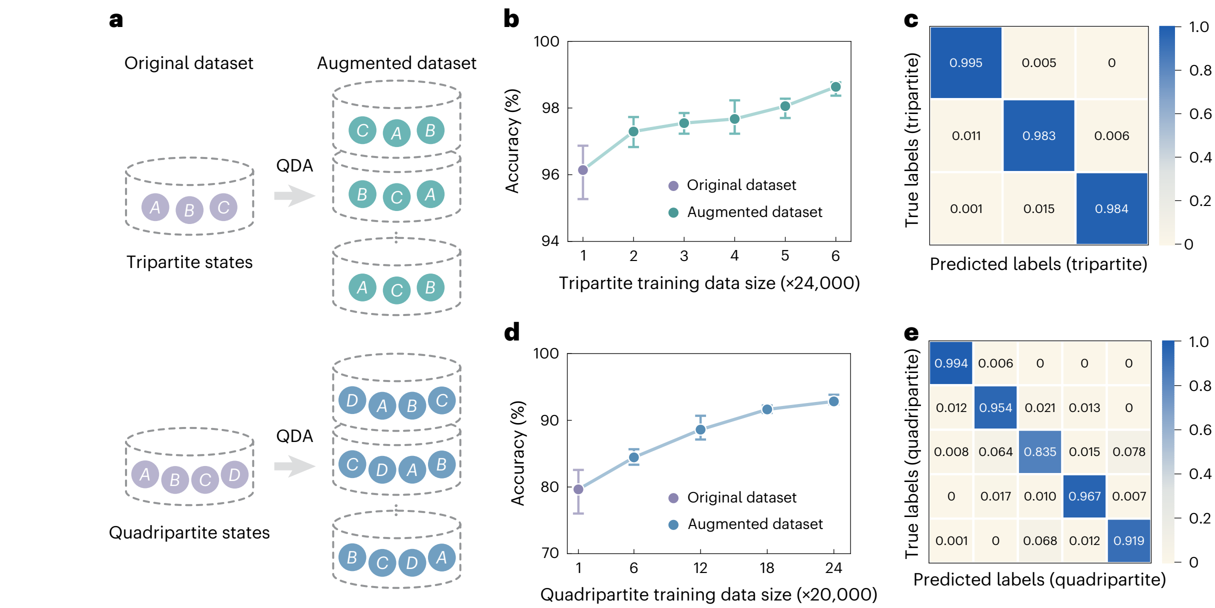

Fig. 3 | Multipartite entanglement structure classification performance after applying QDA. **a**, Schematic of QDA for tripartite (top) and quadripartite (bottom) inputs. **b**, Testing accuracies for tripartite entanglement structure prediction using neural networks trained on the original dataset (24,000 samples, purple marker) and on augmented datasets (green markers). Data are presented as mean values over $n = 4$ independent training runs; error bars indicate the minima and maxima across runs. **c**, Confusion matrix for the tripartite case after QDA with 6× data enlargement. Diagonal entries (top-left to bottom-right) correspond to the prediction accuracies for tripartite partition classes: $1\otimes1\otimes1$, $1\otimes2$, and fully inseparable. **d**, Testing accuracies for quadripartite entanglement structure prediction using neural networks trained on the original dataset (20,000 samples, purple marker) and on augmented datasets (blue markers). Data are presented as mean values over $n = 4$ independent training runs; error bars indicate the minima and maxima across runs. **e**, Confusion matrix for the quadripartite case after QDA with 24× data enlargement. Diagonal entries represent the accuracies for five partition classes: $1\otimes1\otimes1\otimes1$, $1\otimes1\otimes2$, $2\otimes2$, $1\otimes3$ and fully inseparable.

preserving the fully separable partition label $\mathcal{S} = 1\otimes1\otimes1$ (Fig. 2e). Meanwhile, correlation patterns of the convexly combined states can also be efficiently obtained, due to the linearity of the expectation value function. This QDA operation further enlarges our training dataset, and is also valuable for related unsupervised anomaly detection tasks in which separable samples constitute the normal class and entangled samples are treated as anomalies.

Entanglement structure classification

By testing the neural networks on the remaining 20% unseen simulated samples, we assess the classification performance of models trained with the original dataset and augmented datasets, respectively. The accuracy measures the ratio of correctly predicted samples to the total number of samples in the test dataset, that is, $N(\mathcal{S}_{\mathrm{Pred}} = \mathcal{S}_{\mathrm{True}})/N_{\mathrm{total}}$. As a baseline, we apply the QFI-based method to the same homodyne data by first reconstructing the reduced density matrix via maximum likelihood estimation. Due to the limited correlation patterns available for full state reconstruction, this procedure suffers from substantial errors, yielding a classification accuracy of only 0.370 for the tripartite case. We also implement a nearest-neighbour baseline on the input features: for each test correlation pattern, we assign the label of the closest training sample based on Euclidean distance. This non-machine learning baseline achieves accuracies of 0.575 for tripartite states and 0.485 for quadripartite states.

By contrast, neural networks demonstrate strong advantages. When trained on the original dataset only, the network achieves an accuracy of 0.961 for the tripartite case (Fig. 3b, purple markers). After using augmented data in multiples of the original dataset size, the green markers show substantial improvements in the accuracy of entanglement structure prediction. It can be raised to 0.986 when the size of the augmented dataset expands to six times the original size. A confusion matrix is also evaluated to demonstrate the ability of the network to distinguish among different multipartitions (Fig. 3c). In this matrix, each row represents the samples of a true class and sums to 1, whereas each column represents samples of a predicted class. Using our data-augmented network, the entanglement structure in terms of different multipartitions can be detected with a prediction accuracy higher than 0.983 based on limited homodyne measurement data.

It is worth noting that the results presented above are obtained using a combination of both convex combination and mode permutation, with mode permutation contributing 97% of the augmented samples and convex combination of the remaining 3%. As expected, these two operations yield slightly different effects on network performance; an ablation study comparing models trained with only permutation augmentation, only convex-combination augmentation and both of these combined is given in Supplementary Section IIIC.

The QDA method is further tested in the quadripartite entanglement structure classification task. For each quadripartite state, we mix three modes using two beamsplitters and then obtain 16 correlation patterns by measuring the remaining unmixed mode and one of the mixed outputs (Supplementary Fig. 7). A neural network with an architecture similar to that used in the tripartite case is trained with an original dataset containing 20,000 samples. These samples are evenly distributed across five classes of states, each representing a multipartition class of fully separable partition $1\otimes1\otimes1\otimes1$, triseparable partition $1\otimes1\otimes2$, biseparable partitions $2\otimes2$ and $1\otimes3$, and fully inseparable states. As shown in Fig. 3d, the network trained with the original dataset only reaches an accuracy of around 0.796 on 5,000 unseen test samples. After applying QDA, the augmented dataset is expanded 24 times, substantially boosting the accuracy to 0.928 (blue markers). The confusion matrix shown in Fig. 3e highlights QDA improvements in which the augmented network nearly perfectly classifies fully separable states and achieves 0.919 accuracy in identifying fully quadripartite inseparable states. These accuracies can be further improved by feeding in additional correlation patterns, especially for the class $\mathcal{S} = 2\otimes2$, as detailed in the 'Discussion' section. Evidently, with the enhancement

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Table 1 | Neural network performance for different tripartite training dataset sizes

Original dataset size	Accuracy	Augmented dataset size	Accuracy
4,000	0.905	24,000	0.957
8,000	0.910	48,000	0.968
12,000	0.931	72,000	0.972
16,000	0.950	96,000	0.983
20,000	0.958	120,000	0.984
24,000	0.961	144,000	0.986

in QDA, the neural network becomes more powerful in classifying multipartite entanglement structures across all possible multipartitions.

Discussion

Complementing these results, we now examine the performance and adaptability of our network under practical conditions. In particular, we investigate the influence of original training-set size, inclusion of additional correlation patterns, potential information loss due to pattern discretization and statistical fluctuations from finite measurements. We further assess the network's generalization ability to typical experimental noise, as well as to unknown infinite-stellar-rank non-Gaussian states. Finally, we demonstrate the scalability of our approach to five-mode CV systems. These analyses collectively highlight the robustness, flexibility and broader applicability of the data-augmented network for multipartite entanglement classification.

First, we vary the size of the original training set for tripartite states and retrain the network. As illustrated in Table 1, a key observation is that the largest accuracy gap between the original and augmented datasets occurs at 8,000 samples, where the augmented dataset achieves 0.968 compared with 0.910 for the original simulated dataset. However, as the original dataset size increases, the gap narrows, and the accuracy after augmentation converges to 0.986, suggesting that the marginal benefit of augmentation decreases with larger original datasets. This indicates that although DA is beneficial across all dataset sizes, it has a more pronounced impact when the original dataset is smaller.

Second, note that in the 'Results' section, each joint probability distribution of quadripartite states is obtained after mixing three modes; hence, only correlation features for the $A-BCD$, $B-ACD$, $C-ABD$ and $D-ABC$ splittings are captured. However, for states with a biseparable partition $2\otimes 2$, the separability cannot be perfectly represented by these input data. Therefore, the detection accuracy for states with $\mathcal{S}=2\otimes 2$ only reaches 0.835 (Fig. 3e). Even when training is restricted to states with a finite stellar rank, the detection accuracy for $\mathcal{S}=2\otimes 2$ states remains limited to 0.844. To further improve the accuracy of classifying quadripartite entanglement structures, especially for the class $\mathcal{S}=2\otimes 2$, we fed the neural network with new correlation patterns, which are binned from 16 additional joint probability distributions providing correlation features of $A-BC$, $B-AC$, $C-AB$ and $D-AB$ splittings (Supplementary Fig. 8). With these additional distributions, the accuracy for detecting states in the biseparable partition $\mathcal{S}=2\otimes 2$ is improved to 0.922 when the size of the augmented dataset expands to 24 times the original size. Correspondingly, the average accuracy for detecting entanglement structures of quadripartite states is improved to 0.969.

Third, to assess whether the discretization of patterns leads to information loss that could degrade network performance, we apply the t-distributed stochastic neighbour embedding (t-SNE) algorithm to visualize the high-dimensional input correlation patterns. This technique projects each pattern onto a two-dimensional manifold in which the proximity between points reflects the similarity of their underlying correlation structures. As shown in Fig. 4a, the raw, discretized feature space exhibits substantial overlap among samples from different entanglement structure classes. This initial clustering indicates that discretization can indeed lead to near-degenerate representations that are difficult to distinguish via direct observation. After processing by the trained neural network, however, the same dataset becomes markedly well separated. Previously overlapping samples are mapped into distinct, isolated clusters in the learned feature space. These results demonstrate that within the regime considered, the information loss inherent to discretization does not impose a fundamental limitation. Rather, the network's nonlinear feature extraction effectively resolves these near-degeneracies, enabling a robust identification of entanglement structures from experimentally accessible correlation patterns.

Finally, we assess the robustness against statistical fluctuations in correlation patterns, mimicking finite-sample homodyne data in actual experiments. To investigate their impact, we further simulate the limited measurement outcomes via a Monte Carlo sampling method and find that even with finite sampling points, the data-augmented network can still maintain high accuracy.

To be more specific, holding 100 new tripartite samples with statistical fluctuations as the test dataset, we evaluate the neural network's performance. Focusing on states with a finite stellar rank, we denote input samples derived from ideal correlation patterns as $\mathcal{C}_{\text{ideal}}$, and those incorporating statistical fluctuations as $\mathcal{C}_{\text{MC}}$ (Fig. 4b,c, respectively). These fluctuations, arising from homodyne detection sampling, are simulated using a Monte Carlo process (Fig. 4d). Training with 24,000 ideal samples $\{\mathcal{C}_{\text{ideal}}\}$ yields accuracies of 0.929 and 0.976 on the ideal test dataset before and after QDA, respectively. However, when tested on the 100 non-ideal samples $\{\mathcal{C}_{\text{MC}}\}$, the same network architecture achieves accuracies of 0.910 and 0.950 before and after QDA, respectively. Hence, imperfect knowledge of the joint probability distributions only slightly weakens the performance of the neural network. In addition, introducing a small number of samples with statistical fluctuations into the training dataset can help the network learn to recognize and account for the impact of fluctuations on entanglement detection. After using 23,700 ideal samples $\{\mathcal{C}_{\text{ideal}}\}$ and 300 non-ideal samples $\{\mathcal{C}_{\text{MC}}\}$ for training, we still test the neural network on the 100 non-ideal test samples. The network achieves improved accuracies of 0.930 and 0.970 before and after DA, respectively.

To assess the generalization ability, we evaluate our trained network under two challenging conditions beyond its training distribution. First, despite being trained only on simulated data with photon loss, the network maintains high robustness when tested on samples incorporating thermal and phase noise, achieving an accuracy of 0.980 on hundreds of noisy tripartite states (Supplementary Section ID). Second, we test the model on 1,000 unseen non-Gaussian states generated via spontaneous parametric downconversion (Supplementary Fig. 3), which is a physically distinct class of infinite-stellar-rank states absent from the training set. Without fine-tuning the network or using prior information about the states, the network achieves near-perfect accuracy in identifying their entanglement structures. These results demonstrate that our model has learned robust, transferable features of multipartite entanglement that generalize across key experimental imperfections (including photon loss, thermal noise, phase noise and finite sampling) as well as to states beyond the original training distribution.

Beyond feature engineering, a critical consideration for any quantum characterization framework is its scalability to larger systems. For an m-mode system, the complexity of entanglement structures is fundamentally determined by the integer partitions of m. In the five-mode case, this yields seven distinct entanglement classes. To evaluate scalability, we extend our analysis to 7,000 simulated five-mode states (1,000 per class), spanning stellar ranks $r\in\{0,1,2,3,+\infty\}$, with 80% of the data used for training. The input correlation patterns are generated following an $(m-1)$-mode beamsplitter mixing protocol

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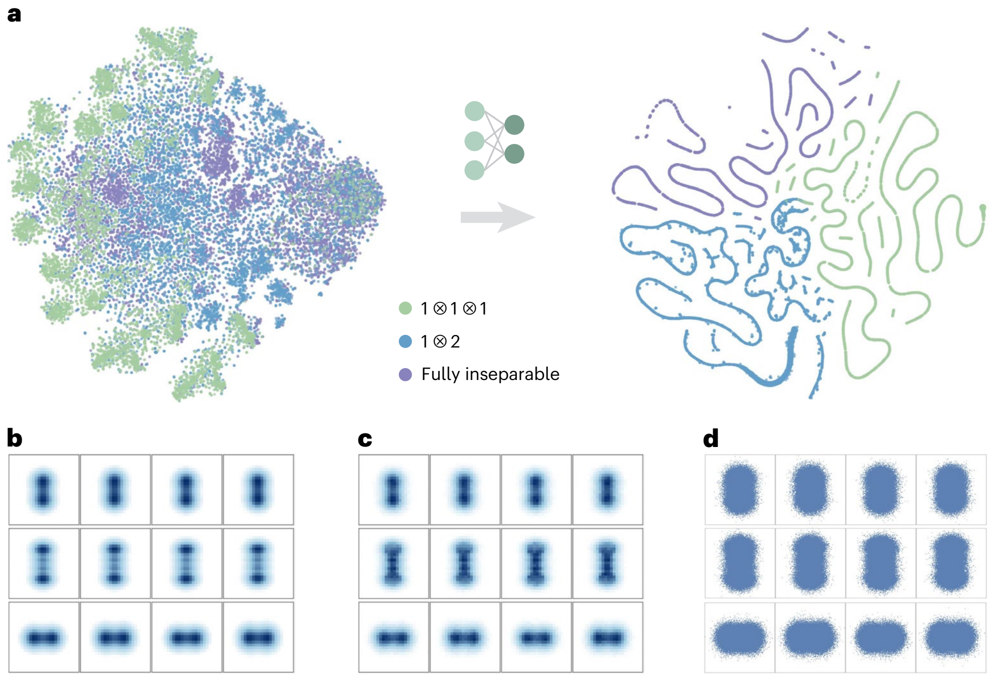

Fig. 4 | Robustness to discretization and statistical fluctuations in correlation patterns. a, Two-dimensional *t*-SNE visualization of tripartite correlation patterns before (left) and after (right) processing by the augmented neural network. The *t*-SNE map of the original discretized correlation patterns shows strong overlap among different multipartition classes, whereas the corresponding projection of the final-layer network features yields clearly separated clusters, suggesting that discretization-induced information loss does not prevent effective classification. **b**, Ideal correlation patterns $\mathcal{C}_{\mathrm{ideal}}$ binned from the exact joint probability distribution. **c**, Noisy correlation patterns $\mathcal{C}_{\mathrm{MC}}$ binned from Monte Carlo sampling. **d**, Monte Carlo sampling of 10^5 statistically independent points for each pattern, simulating a set of joint homodyne detection events.

(Supplementary Fig. 9), and we apply 100-fold augmentation by performing distinct permutations within the training set. The impact of QDA remains substantial as system size increases: classification accuracy for five-mode states improves from 76.7% to 92.4%. As illustrated in the confusion matrix (Fig. 5), several partitions achieve near-perfect identification. Nevertheless, further scaling to larger systems will incur additional computational costs. Generating and labelling diverse CV states, constructing informative features and training larger networks become increasingly expensive as the number of modes and non-Gaussian complexity grow. Hence, substantial further scaling will require a careful management of the overall computational budget.

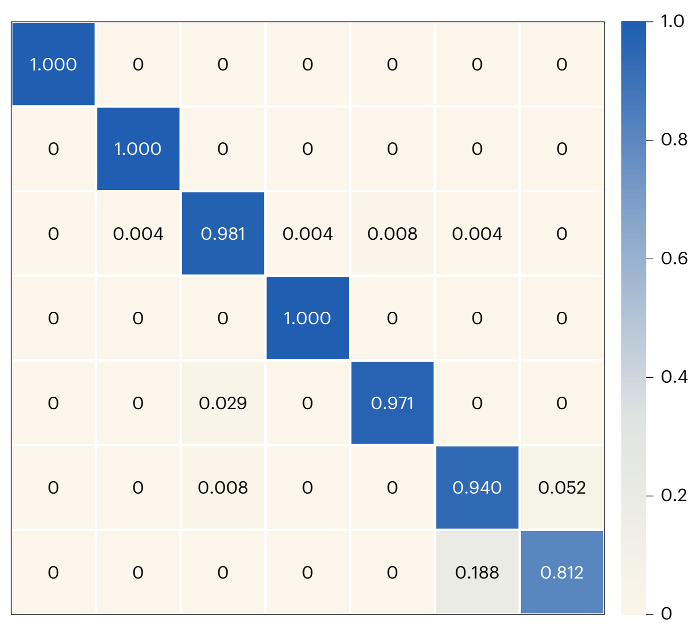

Fig. 5 | Confusion matrix for the five-mode case after QDA with 100-fold data enlargement. The diagonal elements represent prediction accuracies for partitions $1\otimes1\otimes1\otimes1\otimes1$, $1\otimes1\otimes1\otimes2$, $1\otimes2\otimes2$, $1\otimes1\otimes3$, $1\otimes4$, $2\otimes3$ and fully inseparable.

Conclusion

In conclusion, we present a data-augmented deep learning approach for classifying quantum entanglement structures of general multipartite CV states with both finite and infinite stellar ranks, using only feasible homodyne measurement data. This procedure involves training a convolutional neural network in which the network is fed with diverse synthetic datasets generated through the QDA method, rooted in fundamental physical principles. Although the present work focuses on the classification of partition-based entanglement structures, our framework is inherently extensible to the finer-grained discrimination of distinct entanglement types within a given partition. By transitioning from a permutation-invariant mapping to a permutation-equivariant one, the network can effectively distinguish between specific configurations, such as different bipartite splits within the biseparable class (Supplementary Section IIIE).

More broadly, this strategy can be generalized to other complex multipartite scenarios by exploiting the invariance of target quantum properties under specific operations, a concept analogous to 'free operations' in quantum resource theory. Such guidance from resource theory could inform the development of QDA strategies. Future work may fruitfully explore hybrid designs that incorporate this resource invariance, particularly in conjunction with recently developed equivariant neural network architectures[46,47]. Critically, QDA-enhanced networks not only capture the underlying entanglement features but also serve as practically valuable tools for quantum information processing tasks.

Methods

Stellar formalism

To randomly generate m-mode Gaussian and non-Gaussian seed states $\hat{\sigma}_m$, we first characterize them using the stellar formalism[35–37]. In this formalism, we analyse an m-mode normalized pure state $|\psi\rangle$ in terms of its stellar function $F^{\star}_{\psi}(\mathbf{z}) \equiv e^{\frac{1}{2}\|\mathbf{z}\|^2}\langle \mathbf{z}^*|\psi\rangle$, where $|\mathbf{z}\rangle$ is a coherent state with complex amplitude $\mathbf{z}$. The stellar rank r of $|\psi\rangle$ is defined as the number of zeros of $F^{\star}_{\psi}(\mathbf{z})$, representing the minimal non-Gaussian operational cost to engineer the state from the vacuum.

By introducing the notation $\overline{\mathbb{N}} = \mathbb{N} \cup \{+\infty\}$, the stellar hierarchy of CV states is induced by the stellar rank $r \in \overline{\mathbb{N}}$. For instance, $r = 0$ means that the state is Gaussian, whereas $r = 1$ corresponds to a class of non-Gaussian states that contain both single-photon-added and single-photon-subtracted states. Another important and experimentally accessible class of non-Gaussian states, cat states, is of infinite stellar rank $r \to \infty$.

Correlation patterns transformation under mode permutation

For the tripartite quantum system, when exchanging modes A and B, the joint probabilities transform as follows:

$$\begin{array}{c}
\langle X_A; \frac{X_B+X_C}{\sqrt{2}}|\hat{\rho}|X_A; \frac{X_B+X_C}{\sqrt{2}}\rangle \\
\langle \frac{X_A+X_C}{\sqrt{2}}; X_B|\hat{\rho}|\frac{X_A+X_C}{\sqrt{2}}; X_B\rangle \\
\langle \frac{X_A+X_B}{\sqrt{2}}; X_C|\hat{\rho}|\frac{X_A+X_B}{\sqrt{2}}; X_C\rangle \\
\downarrow A \leftrightarrow B \\
\langle X_B; \frac{X_A+X_C}{\sqrt{2}}|\hat{\rho}|X_B; \frac{X_A+X_C}{\sqrt{2}}\rangle \\
\langle \frac{X_B+X_C}{\sqrt{2}}; X_A|\hat{\rho}|\frac{X_B+X_C}{\sqrt{2}}; X_A\rangle \\
\langle \frac{X_B+X_A}{\sqrt{2}}; X_C|\hat{\rho}|\frac{X_B+X_A}{\sqrt{2}}; X_C\rangle.
\end{array} \tag{3}$$

This transformation can be efficiently realized by swapping the first two original probability distributions and transposing them, respectively, leaving the remaining probability distribution unchanged.

For the quadripartite quantum system, when exchanging modes A and B, the joint probabilities will be transformed as below:

$$\begin{array}{c}
\langle X_A; \frac{X_B+X_C+X_D}{\sqrt{3}}|\hat{\rho}|X_A; \frac{X_B+X_C+X_D}{\sqrt{3}}\rangle \\
\langle \frac{X_A+X_C+X_D}{\sqrt{3}}; X_B|\hat{\rho}|\frac{X_A+X_C+X_D}{\sqrt{3}}; X_B\rangle \\
\langle \frac{X_A+X_B+X_D}{\sqrt{3}}; X_C|\hat{\rho}|\frac{X_A+X_B+X_D}{\sqrt{3}}; X_C\rangle \\
\langle \frac{X_A+X_B+X_C}{\sqrt{3}}; X_D|\hat{\rho}|\frac{X_A+X_B+X_C}{\sqrt{3}}; X_D\rangle \\
\downarrow A \leftrightarrow B \\
\langle X_B; \frac{X_A+X_C+X_D}{\sqrt{3}}|\hat{\rho}|X_B; \frac{X_A+X_C+X_D}{\sqrt{3}}\rangle \\
\langle \frac{X_B+X_C+X_D}{\sqrt{3}}; X_A|\hat{\rho}|\frac{X_B+X_C+X_D}{\sqrt{3}}; X_A\rangle \\
\langle \frac{X_B+X_A+X_D}{\sqrt{3}}; X_C|\hat{\rho}|\frac{X_B+X_A+X_D}{\sqrt{3}}; X_C\rangle \\
\langle \frac{X_B+X_A+X_C}{\sqrt{3}}; X_D|\hat{\rho}|\frac{X_B+X_A+X_C}{\sqrt{3}}; X_D\rangle.
\end{array} \tag{4}$$

This transformation can also be efficiently realized by swapping the first two original probability distributions and transposing them, respectively, and leaving the other two probability distributions unchanged.

For the additional quadripartite joint probability distributions, the transformation is similar to that in equation (3). The same transformation can be applied to the correlation patterns, which are binned from these joint probability distributions. Correspondingly, the correlation patterns are also flipped and exchanged according to the above transformations. Although index swapping does not change the characterization of the quantum states, the rearranged correlation patterns are treated as new samples by the neural network. Therefore, the transformation results in a modified sample that can still be assigned to the original entanglement structure label $\mathcal{S}$.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The datasets generated and analysed in this study are available via Code Ocean at https://doi.org/10.24433/CO.9492084.v2 (ref. 48). Source data are provided with this paper.

Code availability

The code used for QDA, neural network training and model evaluation is available via Code Ocean at https://doi.org/10.24433/CO.9492084.v2 (ref. 48). The corresponding GitHub repository is available at https://github.com/XiaotingGaoPhys/Multipartite-Entanglement-Classification-QDA (ref. 49) and has been archived on Zenodo at https://doi.org/10.5281/zenodo.20928779 (ref. 50). The neural networks were implemented using TensorFlow[51].

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Author contributions

X.G., Y.X. and Q.H. conceived the original idea. Q.H. managed the project. X.G. performed the numerical simulations and developed the neural network model. M.T. proposed the optimized QFI criterion algorithm to label the quantum states. All authors discussed the results and contributed to the final version of the paper.

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Competing interests

The authors declare no competing interests.

Additional information

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